

Token Averaging for Compute-Efficient Language Model Pretraining

Working draft: results are being added as experiments complete

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July 2026

Abstract

Training a language model on **more text costs** more compute: a transformer spends one sequence position and one output prediction per token, so doubling the training data roughly doubles the work. We study *token averaging*, which makes **each token cheaper**. Before the first attention layer we replace every k consecutive token embeddings by their mean, so the transformer runs on T/k positions instead of T and predicts the first token of the next group at each position. Processing a given amount of text then costs about $1/k$ as much, and we spend that saving on $k\times$ more training data. At that point the averaged run executes the same number of positions and predictions as the baseline, and its shorter sequences make attention cheaper, so it finishes at $0.93\times$ the baseline’s training FLOPs at 50M and 125M parameters, $0.94\times$ at 250M and $0.95\times$ at 500M. Because a k -averaged model predicts only one token per group, its loss and the baseline’s are not measurements of the same quantity, so we make comparability part of the method. We train with a **randomly placed group boundary at each step**, which gives every token position an equal share of the training signal. **We then compare across k with an *offset-ensemble* evaluation: run the averaged model k times per sequence with the group boundaries shifted by $0, 1, \dots, k-1$ tokens, so that every position is predicted exactly once.** Each pass is over a sequence k times shorter, so the k passes together cost about one baseline forward pass. On this metric a 125M model with $k=2$ reaches 3.4% lower perplexity than its baseline while using $0.93\times$ the training FLOPs. **Reading each family’s loss-versus-compute curve at equal loss, the baseline needs $1.48\times$ more compute at 50M, $1.43\times$ at 125M and $1.31\times$ at 250M to reach the averaged model’s loss.** This saving shrinks at every scale, and at 500M it falls into our measurement noise: the averaged model still ends ahead on the comparable metric, but by 0.002 nats, **an order of magnitude less than the spread we get from changing the training protocol alone.** **We therefore report the 500M pair as a decay to parity rather than a win.** **A set of controlled 50M runs shows that the remaining gain comes from processing data more cheaply, not from the longer raw context that averaging also makes affordable: spending the saving on longer context instead of more data is worse on packed web text at these scales, while averaging matches a full-attention model at the same context for $1.14\times$ less training compute and half the inference KV cache.** Finally, comparing pooling functions at 50M under one matched protocol, weighting each token by both its content and its position within the group ($d_{\text{model}}+1+k$ extra parameters) gives the lowest native loss we measured, beating the mean by 0.037 nats at $k=2$ and 0.283 nats at $k=4$. [TODO: add 1B endpoints; add full-position numbers at 50M/250M; add 500M seed replicates; tighten to venue length]

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1 Introduction

Large language models are trained by predicting the next token over very large corpora, and that training is expensive. A transformer processes one sequence position per token and makes one output prediction at each position, so doubling the amount of text roughly doubles both counts; attention adds a further cost that grows with sequence length. Data volume is not a secondary concern within that budget. Compute-optimal scaling analyses (Hoffmann et al., 2022) estimate that the best parameter count and the best training-token count both grow as roughly $C^{1/2}$, so about half of any extra compute should go to seeing more data. Making each token of data cheaper to train on is therefore worth trying, provided that what the model loses per token is smaller than what the extra data gains. Whether that trade is favourable is an empirical question, and it is the question this paper measures.

We study a minimal mechanism of this kind: *token averaging*. Before the first transformer layer we split the input into non-overlapping groups of k consecutive tokens and replace each group’s k embedding vectors by their mean. The transformer then runs on T/k positions instead of T , and at each position it predicts the first token of the following group. Nothing else changes: the architecture, tokenizer, data source, and optimizer are the baseline’s, and mean pooling adds no parameters.

Because k raw tokens now share one transformer position and one output prediction, running the model over a given amount of text costs about $1/k$ of what the baseline costs. That is the saving this paper is about, and we spend all of it on data: at the same compute budget the averaged model trains on $k\times$ more raw tokens. Doing so puts the position count and the prediction count back to exactly the baseline’s totals, but the averaged run still ends *below* the baseline’s FLOPs, because its transformer sequences are k times shorter and attention therefore costs less ($0.928\times$ at our 50M configuration, $0.932\times$ at 125M, $0.940\times$ at 250M, $0.946\times$ at 500M; the ratio approaches 1 as d_{model} grows and attention becomes a smaller share of the cost, Section 3.5). Averaging is lossy, so the model learns less from each token. We measure whether that loss is smaller than the gain from seeing $k\times$ more data, under one metric and one compute budget.

Comparing an averaged model to a standard one takes care, so we build what is needed into the method. A k -averaged model assigns probabilities to only one token per group, so its loss and the baseline’s are averages over different sets of predictions and are not measurements of the same quantity. We address this in two ways. First, we train with a *random group offset*: at each step the group boundaries are shifted by a random number of tokens, so that over training every token position is predicted equally often (Section 3.3). Second, every comparison across k uses an *offset-ensemble* evaluation: we run the model k times over each evaluation sequence with the group boundaries shifted by $o = 0, \dots, k-1$ tokens, so that every position past the first group is predicted exactly once from its full compressed prefix (Section 3.6). This gives a true full-sequence negative log-likelihood, and because each of the k passes is over a sequence k times shorter, together they cost about one baseline forward pass. The same idea gives a way to generate text one token at a time (Appendix B).

The saving can be spent in two ways, and we separate them. Our main setting, which we call the **more-data setting**, keeps the raw sequence length at the baseline’s 1024 tokens: the transformer runs on $1024/k$ positions, every prediction still sees 1024 raw tokens of context, and the freed compute buys $k\times$ the training data. This is the setting we scale. To find out where its gains come from we run two further arms. The **more-context setting** keeps the transformer length at the baseline’s 1024 and spends the whole saving on $k\times$ longer raw context instead of more data. A standard model with a genuinely widened attention window then tells us whether any effect of that extra context is specific to averaging (Section 3.4).

Findings. Our main experimental findings, on FineWeb with GPT-NeoX-style models trained from scratch, are as follows.

1. **At small scale, averaging wins at equal compute on a strictly comparable metric.** We score the 125M pair with the offset-ensemble protocol, so both models are evaluated on identical sequences and identical target positions (3,409,119 predictions each). The $k=2$ model reaches 4.141 nats/token against 4.177 for the baseline: 3.4% lower perplexity at $0.93\times$ the baseline’s training FLOPs. The two passes of the ensemble score almost equally well (4.145 with the group boundaries at offset 0, 4.138 at offset 1), so where the boundaries fall carries little information on packed data (Section 5.2). Reading each family’s loss-versus-compute curve at equal loss, the baseline needs $1.48\times$ more compute at 50M, $1.43\times$ at 125M and $1.31\times$ at 250M to reach the averaged model’s loss. Those three multipliers come from native losses and are provisional until we rescore 50M and 250M with the offset ensemble (Section 5.1). [TODO: add 50M/250M full-position numbers]
2. **The efficiency multiplier shrinks as models grow and reaches our noise floor at 500M.** The equal-loss saving falls at every one of the first three scales, from $1.48\times$ at 50M to $1.31\times$ at 250M, and the endpoint loss gap narrows with it ($0.074 \rightarrow 0.033 \rightarrow 0.011$ nats). At 500M that gap is 0.002 nats on the comparable metric. It is still in the averaged model’s favour, but it is an order of magnitude below the 0.070-nat spread that the training protocol alone produces at 50M, and it is too small for the equal-loss multiplier to be measured stably: two adjacent evaluation points of the same run put the multiplier anywhere from $1.08\times$ to unattainable. We therefore report this as a decay to parity within noise rather than as a fourth win, and we do not extrapolate a crossover scale in either direction, because such a projection moves by more than a factor of two under bookkeeping choices that leave every measured loss unchanged (Section 5.1). Whether a small positive effect survives at and beyond this scale is what the 500M seed replicates and the 1B arm are meant to answer.
3. **The gain comes from cheaper data, not from longer context.** Averaging makes longer raw context affordable as well as more data, so we trained four 50M models to separate the two: the baseline, a standard model with a genuinely doubled attention window, and the more-context setting at $k=2$ and $k=4$. Longer raw context hurts in every arm on packed FineWeb. Widening a standard model’s attention window from 1024 to 2048 tokens costs +0.214 nats and $1.14\times$ the compute, and spending the averaging saving on longer context instead of more data is worse at both $k=2$ and $k=4$. On this data, then, what averaging buys is data per FLOP (Section 5.3).
4. **When longer context is what you want, averaging is the cheaper way to get it.** The $k=2$ averaged model and the full-attention model, both seeing 2048 raw tokens of context, finish at the same loss. The averaged route spends $1.14\times$ less training compute and holds half the KV cache at inference (Section 5.3). [TODO: full-position endpoints for the two 2048-context arms]
5. **Weighting tokens by content and position pools better than taking the mean, at both compression ratios.** We trained six pooling functions at 50M under one matched protocol. The best was a small scorer ($d_{\text{model}}+1+k$ parameters) that weights each token by its embedding and by which slot of the group it occupies; it gave the lowest loss at both $k=2$ (4.326, against 4.363 for the mean) and $k=4$ (4.163, against 4.446). Each pooling function was trained once, so the 0.037-nat margin at $k=2$ is an observed ordering rather than an

established one. The other runs suggest why position matters. At $k=4$, a fixed rule that simply weights later tokens of the group more heavily¹ recovers most of the gap (4.265), while a scorer that sees only token content and not position² recovers none of it (4.445). What the mean discards therefore looks like position information rather than content. One confound is that the $k=4$ position-aware scorer starts from that fixed recency rule while the content-only scorer starts from a uniform mean (Section 5.4).

Contributions. (1) A compute-controlled study of token averaging as a *permanent* operating point rather than a warm-up phase, across four model scales, with budgets set so the two arms of each pair end at equal compute. (2) A way to make patch-style models comparable to ordinary ones: random-offset training, the offset-ensemble evaluation that puts both on the same full-sequence likelihood, and a rotating-offset design for generation. (3) Evidence that longer raw context hurts on packed web text at these scales, so the benefit of averaging is cheaper data, and that when context is what you want, averaging is the cheaper way to get it. (4) A matched-protocol comparison of pooling functions at 50M in which weighting by content and position gives the lowest loss at both compression ratios, pointing to position within the group as the ingredient the mean discards.

2 Related Work

[TODO: This section is deliberately deferred. Anchor points to cover when we write it:]

- **Patch-level training.** Shao et al. (2024; ICLR’25), “Patch-Level Training for Large Language Models”: the closest prior work. It averages K token embeddings into patches and predicts the next patch, but uses patch-level training only as a *warm-up* followed by token-level fine-tuning, at 370M–2.7B scale. Our deltas: patch-level as the final operating point (made comparable by the offset-ensemble evaluation), FLOP-controlled scaling-law treatment rather than single budget points, and the context-extension decomposition.
- **Hierarchical / byte-level patching.** MegaByte, Byte Latent Transformer (dynamic patch boundaries), Block Transformer, Funnel-Transformer, Hourglass. These change the architecture to compress and re-expand; we deliberately keep the architecture untouched.
- **Multi-token prediction.** Gloeckle et al. 2024; DeepSeek-V3’s MTP objective. These predict several targets from uncompressed inputs; we compress the inputs instead.
- **Context compression.** Gist tokens, AutoCompressor, ICAE: these compress *prompts* at inference; we compress *all* input during pretraining.
- **Scaling laws.** Hoffmann et al. 2022 (Chinchilla) for the compute-optimal framing and the $D^* \approx 20N$ heuristic we use to set budgets; Kaplan et al. 2020.
- **Tokenizer granularity.** Work on tokenizer vocabulary size and bits-per-byte comparisons, relevant to the coarser-tokenizer control we plan (Section 6).

¹The exponential recency profile of Section 3.7: weights rise geometrically across the group so the last token weighs 20 times the first. It has no learned parameters.

²A single linear scorer applied to each embedding independently. It runs before the network has any positional information, so it can react to what a token is but not to where in the group it sits.

3 Method

3.1 Token averaging

Let $t_1, \dots, t_T$ be a sequence of raw tokens and $e_i \in \mathbb{R}^{d_{\text{model}}}$ their learned input embeddings. For a fixed averaging factor k , we split the sequence into $n = \lfloor T/k \rfloor$ non-overlapping groups of k consecutive tokens and replace each group by the mean of its embeddings:

$$\bar{e}_j = \frac{1}{k} \sum_{i=(j-1)k+1}^{jk} e_i, \quad j = 1, \dots, n. \quad (1)$$

The transformer body (attention layers, feed-forward layers, RoPE, final norm) runs unchanged on the compressed sequence $\bar{e}_1, \dots, \bar{e}_n$ and produces n output states, and the output head produces logits over the vocabulary at each of them. At $k=1$ the method is exactly a standard language model. Mean pooling adds *no new parameters*; Section 3.7 describes the alternative pooling functions we compare, the largest of which adds 517.

One point about position is worth noting. Rotary position embeddings are applied inside the transformer to the *compressed* positions $1, \dots, n$, so neighbouring transformer positions are always k raw tokens apart. The compressed sequence therefore looks positionally the same no matter where the group boundaries fall in the raw text, which matters for the offset analysis in Section 3.6.

3.2 Training objective

We index groups and positions from 1. Group j covers the raw tokens $t_{(j-1)k+1}, \dots, t_{jk}$, and sits at compressed position j of the transformer. At that position the model predicts the *first token of the next group*, which is t_{jk+1} : it has seen the first jk raw tokens in compressed form, and must name the token that comes next. The table below shows the $k=2$ case on eight raw tokens, where e_i is the embedding of raw token t_i .

raw tokens	t_1	t_2	t_3	t_4	t_5	t_6	t_7	t_8
transformer input	$\bar{e}_1 = \frac{e_1+e_2}{2}$		$\bar{e}_2 = \frac{e_3+e_4}{2}$		$\bar{e}_3 = \frac{e_5+e_6}{2}$		$\bar{e}_4 = \frac{e_7+e_8}{2}$	
target	t_3		t_5		t_7		(none)	

The loss is ordinary cross-entropy, averaged over all predicted positions. Two properties of this objective drive much of the paper. First, the model predicts only the tokens $t_{k+1}, t_{2k+1}, t_{3k+1}, \dots$, one per group. It never assigns a probability to any other token, which is the root of the evaluation problem in Section 3.6. Second, each prediction is made from a lossy view of the past: the most recent group reaches the model only through its mean, so the order of the tokens inside that group is lost. Section 5.4 measures what that costs, and whether other pooling functions recover it.

3.3 Random-offset training

If the groups always started at the beginning of the sequence, the model would only ever be asked to predict the raw positions $k+1, 2k+1, 3k+1, \dots$, and would never be asked about any of the others. We remove that asymmetry directly. At each training step we draw an offset $o \sim \text{Uniform}\{0, \dots, k-1\}$, drop the first o tokens of the batch, and then group as usual. The groups now cover $(o+1 \dots o+k)$, $(o+k+1 \dots o+2k)$, and so on, and the predicted positions become $o+k+1, o+2k+1, \dots$. In other words, the whole pattern of predictions slides along the text by o

tokens. Since o takes each of its k values about equally often, every token position ends up being predicted in roughly the same number of steps.

Sequence packing already randomizes where the group boundaries fall *relative to the underlying documents* (Section 3.6); random offsets additionally make the training signal uniform across positions. At evaluation the offset is fixed to 0, or swept, as in the offset-ensemble metric. The only cost is the $o \leq k-1$ tokens we drop: they shorten each sequence by less than one group, so the batch loses at most one prediction per sequence. Nothing else about training changes.

3.4 Two ways to spend the saving: more data or more context

The compute that averaging frees up can be spent on more data or on longer raw context. We use the first as the paper’s method and the second as an ablation that shows where the first one’s gains come from. Throughout, `seq_len` is the number of *raw* tokens per training sequence, and $L = \text{seq_len}/k$ is the number of positions the transformer actually processes.

More data (the paper’s main setting). The raw `seq_len` stays at the baseline’s 1024. The transformer length falls to $L = 1024/k$, so each raw token costs about $1/k$ of baseline, and the model trains on $k \times$ the raw tokens for the same total compute. Every prediction still sees 1024 raw tokens of context; only the way that context is represented changes.

More context (ablation). Raise `seq_len` to $k \times 1024$, so that $L = 1024$ matches the baseline’s transformer length. Each sequence now packs $k \times$ the raw context into the same amount of transformer computation. The cost per raw token is again $1/k$ of baseline, so this setting also reaches equal total compute at $k \times$ the tokens.

The two settings use identical models, data, and optimization; they differ only in what the freed compute buys. A third arm tells us whether any effect of the extra context is specific to averaging: a standard $k=1$ model whose attention window is genuinely widened to 2048 tokens (`seq_len` = L = 2048). This model gets the same 2048 tokens of context the ordinary way, and it pays for them. Doubling the window doubles the attention term in the cost model of Section 3.5, which at $d_{\text{model}}=512$ raises the cost per raw token to $1.14 \times$ the baseline’s.

3.5 FLOPs accounting

We compute all FLOPs from first principles. For a transformer with hidden size d , n_{layers} layers, and transformer sequence length L , our architecture (attention with Q, K, V and output projections; SwiGLU FFN with expansion factor 2.5) costs $23d^2 + 4Ld$ multiply–accumulate FLOPs forward per layer per position: $8d^2$ for the four attention projections, $15d^2$ for the SwiGLU FFN, and $4Ld$ for the attention scores and the value aggregation (Appendix A derives these). Each position also produces one next-token prediction through the dense vocabulary projection, at $2dV$ FLOPs forward for the padded vocabulary size $V = 50,304$. Multiplying by the usual $3 \times$ for forward plus backward, a run that consumes D raw tokens at raw sequence length `seq_len` costs

$$\text{FLOPs}(D) = \frac{3D}{k} \left[n_{\text{layers}} (23d^2 + 4Ld) + 2dV \right], \quad L = \text{seq_len}/k. \quad (2)$$

We include the output projection because at these scales it is a large part of the bill and not a rounding error: it is 44.2%, 27.8%, 18.5% and 12.0% of the baseline’s total training FLOPs at 50M, 125M, 250M and 500M respectively, since $V \approx 98d$ at $d = 512$ and still $V \approx 39d$ at $d = 1280$.

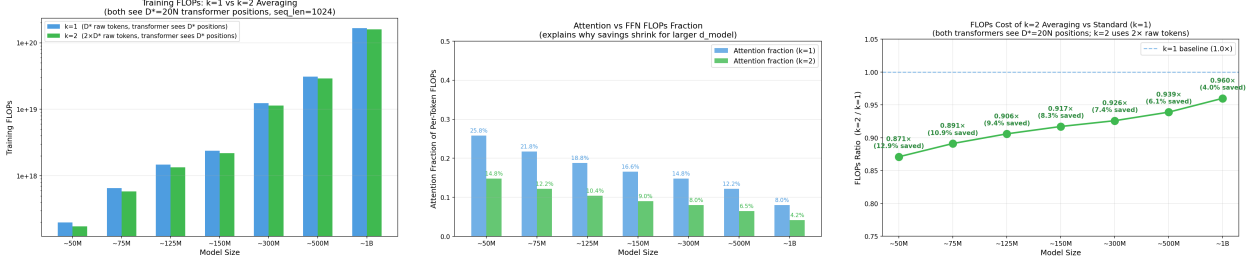


Figure 1: Structure of the transformer-body FLOPs across model widths (the output projection of Eq. (2) is excluded from these three panels, which are about the body’s internal composition). **Left:** absolute body FLOPs per raw token. **Middle:** the attention share of per-position body compute shrinks with d_{model} . **Right:** the per-token body FLOPs ratio of $k=2$ averaging to baseline at $2\times$ the data, across widths. The $k\times$ data throughput does not depend on width; the attention saving is what remains at equal compute, and it is largest at small widths, which is why the final ratio in Table 2 rises towards 1 with scale.

Equation (2) makes the arithmetic of the more-data setting clear. Two comparisons are easy to blur, so we separate them.

At a fixed number of raw tokens D , averaging processes D/k positions instead of D . The two terms that are linear in the position count, the $23d^2$ per layer and the $2dV$ head, therefore fall by a factor k per raw token. The attention term falls by k^2 : one factor k from having fewer positions, and another because each of those positions attends over a sequence k times shorter.

At equal total compute, the averaged run converts that saving into kD raw tokens. The two linear terms then return *exactly* to the baseline’s totals, because the averaged arm now runs the same number of transformer positions and makes the same number of predictions as the baseline. All that is left of its advantage is the attention term, which is still lower because each position attends over L/k positions rather than L . **This sets a ceiling on the method under our accounting, and it is worth stating plainly: at equal compute the averaged arm’s final cost is $0.928\times$ the baseline at 50M, $0.932\times$ at 125M, $0.940\times$ at 250M and $0.946\times$ at 500M, and the ratio rises towards 1 with scale because attention is a shrinking share of the per-position cost as d grows at fixed L .** So what token averaging really buys is *data throughput*: $k\times$ more raw data flows through the same number of positions, with a modest attention saving on top. What that throughput is worth in loss is the empirical question.

3.6 The evaluation problem and the offset-ensemble fix

The problem. The number that defines a language model is the full-sequence log-likelihood $\sum_i \log p(t_i | t_{<i})$. The $k=1$ baseline’s eval loss is the mean of all $T-1$ of those terms per sequence. A k -averaged model’s eval loss, as trained, is the mean over only the $\sim T/k$ positions it predicts, because it has no probability to report for the rest. The two numbers are therefore averages over *different sets of terms*, and more evaluation data does not fix that: evaluating the averaged model on $k\times$ as many batches matches the number of predictions, not the task. The same limitation affects generation: a model that predicts one token per group cannot decode one token at a time as trained.

The fix. At evaluation time, run the model k times over each sequence, shifting the group boundaries by $o = 0, 1, \dots, k-1$ raw tokens. At offset o the groups cover tokens $(o+1 \dots o+k)$,

$(o+k+1 \dots o+2k)$, and so on, and the model’s targets are the raw positions $o+jk+1$ for $j = 1, 2, \dots$, that is $t_{o+k+1}, t_{o+2k+1}, \dots$. Across the k offsets these target sets never overlap, and together they cover every position past the first group, each predicted exactly once from its full compressed prefix. The result is a genuine full-sequence **NLL** over all scored positions, directly comparable to the baseline’s on the same positions. It costs k forward passes over sequences of length $L = T/k$, so about *one* baseline forward pass in total. We also score the baseline on exactly this set of positions, so that the two numbers cover the same predictions and not merely the same number of them. The same idea gives a decoding scheme: while generating, rotate the offset so that a group boundary always ends at the current position. This needs k KV caches, one per offset, but each is $1/k$ the length, so the total cost is about that of one ordinary decoder (Appendix B).

Why every offset works about as well. Random-offset training (Section 3.3) makes the model see all k offsets while training. Two further facts mean it would largely handle them anyway. First, training sequences are cut from a continuous token stream into fixed-length chunks, and those cuts fall at arbitrary points inside documents, so the group boundaries are already uniformly random *relative to the text* whichever offset we use. Second, RoPE acts on compressed positions (Section 3.1), so the positional geometry the transformer sees is the same at every offset. The measured per-offset losses bear this out: they are essentially equal (Section 5.2).

3.7 Pooling functions

Everything the transformer sees passes through the pooling function, so we vary it while holding everything else fixed. Each variant below compresses a group of embeddings into one vector and leaves the rest of the pipeline untouched.

Mean (default). Equal weights $1/k$. No parameters.

Fixed recency weights. $\bar{e} = \sum_i w_i e_i$ with $w_i \propto \exp(\alpha i)$, where α is chosen so the last token of a group weighs 20 times the first, then L_1 -normalized. This is a pure recency rule with no learned parameters: the token closest to the prediction target dominates. (We also implemented linear, Gaussian, and triangular profiles; the exponential one is the version we trained.)

Learned from content only. A single shared linear scorer $s_i = w^\top e_i + b$ gives one number per token. The weights within a group are $\alpha = \text{softmax}(s_{1..k})$, and $\bar{e} = \sum_i \alpha_i e_i$. This adds $d_{\text{model}} + 1 = 513$ parameters, initialized at zero so training starts exactly at mean pooling. The scorer sees each embedding on its own, *before* the network has any positional information, so it can react to what a token is but not to where in the group it sits. That turns out to matter (Section 5.4).

Learned from content and position. The same scorer plus one learned bias per slot of the group: $s_i = w^\top e_i + b + p_i$, with $p \in \mathbb{R}^k$ trained jointly, for $d_{\text{model}} + 1 + k$ parameters. We initialize p uniform at $k=2$ and at the exponential recency profile at $k=4$, which is the best fixed rule we found at each compression ratio, so the model starts from the strongest prior we know of and can then adapt both terms.

Overlapping groups. Average groups of $w=4$ tokens with stride $s=2$. This gives the same $2\times$ compression as $k=2$, but neighbouring groups share tokens, so no token sits on a hard boundary. Targets are the first token past each group.

Word-aligned groups. Groups of variable size (at least k tokens, force-closed at $2k$) that end only at word boundaries, detected from the tokenizer’s whitespace-prefix convention, so no group straddles two words. On FineWeb the realized compression is $\sim 2.1\times$ at $k=2$, slightly above nominal, and the predicted positions are word starts. Both facts affect comparability, and we handle them in Appendix C.

4 Experimental Setup

Architecture. All models are GPT-NeoX-style decoder-only transformers built from the OLM implementation: pre-norm blocks, rotary position embeddings, SwiGLU FFN with expansion factor 2.5, and a padded vocabulary of 50,304 (tokenizer vocabulary 50,257). Table 1 lists the four scales. Input and output embeddings are tied at all scales.

Table 1: Model configurations. “Raw tokens” are the training budgets for the $k=1$ / $k=2$ arms, set so both arms end at approximately equal total training FLOPs (Eq. 2); the $k=1$ budgets follow the Chinchilla $D^* \approx 20N$ heuristic. All main runs use raw `seq_len` = 1024, the more-data setting.

Scale	Params	d_{model}	Heads	Layers	LR	Raw tokens ($k=1$ / $k=2$)
50M	50.9M	512	8	8	2.0×10^{-4}	1B / 2B
125M	123.5M	768	12	12	1.6×10^{-4}	2.5B / 5B
250M	252.8M	1024	16	16	1.4×10^{-4}	5B / 10B
500M	479.3M	1280	20	22	1.2×10^{-4}	10B / 20B

Data and tokenization. All models train from scratch on FineWeb, tokenized with the EleutherAI Pythia-70m (GPT-NeoX) BPE tokenizer. Runs up to and including 250M draw from `sample-10BT`. The 500M pair does not fit in that slice, because its $k=2$ arm alone consumes 20B raw tokens, so both 500M arms draw from `sample-100BT` instead. We change the slice for both arms of the pair, so it does not affect the comparison between them, but it does mean the 500M scale point is not drawn from exactly the same subsample as the smaller ones. The token stream is packed into fixed-length sequences without document-boundary masking. That is standard practice, and it matters for the context experiments (Section 5.3). Evaluation uses a held-out split of the same distribution, identical across all arms of a comparison.

Optimization. AdamW with a cosine learning-rate schedule: 2,000 warmup steps, decay to 10% of peak over exactly the run’s total step count. Peak learning rates follow Table 1. Mixed-precision (bfloat16) training on NVIDIA GPUs via PyTorch DDP; gradient checkpointing at 250M but not at 500M, where memory-efficient attention made it unnecessary and it measurably reduced throughput. Batch sizes and step counts per run are tabulated in Appendix E.

Budgets and equal-compute design. At each scale the $k=1$ arm trains on about $20N$ raw tokens and the $k=2$ arm on twice that. By Eq. (2) this puts the two endpoints at about equal training FLOPs: exactly equal in the two linear terms, and in fact slightly below the baseline for the $k=2$ arm because its attention term is smaller ($0.928\times$ at 50M, $0.932\times$ at 125M, $0.940\times$ at 250M, $0.946\times$ at 500M). The ratio rises towards 1 with scale, because attention is a shrinking fraction of the per-position cost as d_{model} grows at fixed L . The free part of the saving therefore erodes with scale, and the comparison gets stricter as the models get bigger.

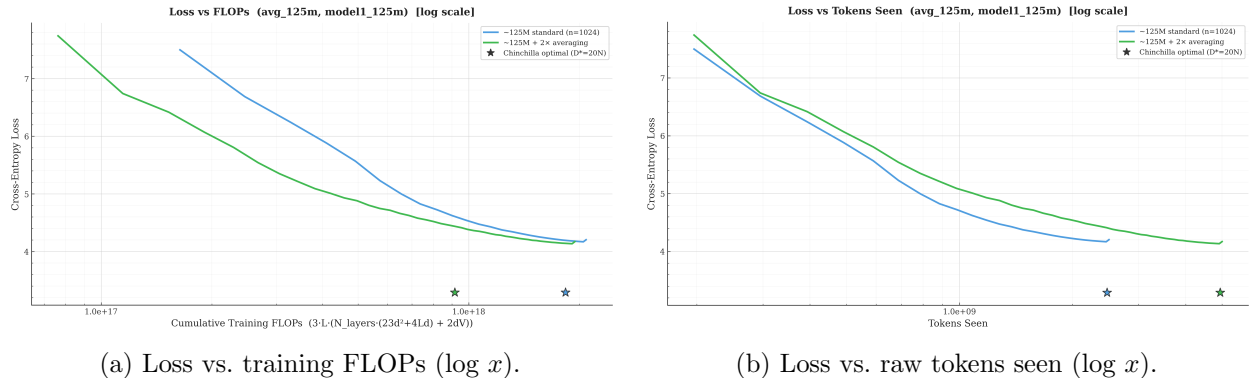


Figure 2: 125M comparison, native eval losses. On the FLOPs axis (left) the $k=2$ curve sits below the baseline curve along the whole trajectory at equal FLOPs; the endpoint comparison on the full-position metric is in Section 5.2. On the raw-token axis (right) the averaged model learns slightly *less* per token (averaging is lossy), but each of its tokens costs roughly half the compute. Stars mark Chinchilla-optimal reference points ($D^*=20N$).

We read every curve two ways: vertically, at equal compute, comparing loss; and horizontally, at equal loss, comparing compute. Both are needed because late-training loss curves are very flat, so a small vertical gap corresponds to a large horizontal one.

Metrics. Two metrics appear in this paper, and each has a fixed job. **Native eval loss** is cross-entropy in nats per predicted token on held-out data, at whatever positions the model natively predicts (offset 0). It is what training logs, and we use it for training curves and for comparing runs that share the same k and grouping scheme. **Full-position NLL** is the offset-ensemble metric of Section 3.6. It scores every position of the identical evaluation sequences, and it is the metric for every comparison across k . We evaluate all final checkpoints under both. [TODO: fill in full-position numbers for the 50M and 250M pairs and the context arms as runs complete]

5 Results

5.1 Equal-compute comparison: an advantage that shrinks with scale

This subsection reports the training curves and **final losses of the main more-data pairs**. The full-position comparison, which is the metric behind every conclusion we draw across k , follows in Section 5.2.

Figure 2 shows the 125M pair; Table 2 summarizes endpoints at all scales. Three observations.

First, on each arm’s own native metric the averaged model ends below the baseline at 50M, 125M and 250M, while spending fewer FLOPs to get there ($0.928\times$ at 50M, $0.932\times$ at 125M, $0.940\times$ at 250M, $0.946\times$ at 500M). The vertical gaps are modest, and they shrink steadily as the models grow: 0.074 nats at 50M, 0.033 at 125M, 0.011 at 250M, and -0.003 at 500M, where the ordering on this metric flips. The horizontal reading is the one that matters in practice. At 125M the baseline needs its full 2.08×10^{18} FLOPs to reach loss 4.204, while the averaged model reaches that loss at 1.46×10^{18} , a $1.43\times$ compute saving. The same reading gives $1.48\times$ at 50M and $1.31\times$ at 250M.

Those flat curves are also why we quote no multiplier at 500M. Once the vertical gap falls to a few thousandths of a nat, the FLOPs value at which one curve crosses the other’s final loss is set by noise rather than by anything stable about the two families. Reading the 500M curves at their last

Table 2: Main equal-compute comparison (more-data setting, raw `seq_len` 1024). Losses are final native eval CE (nats/token) at each model’s own prediction positions; the comparison across k on the full-position metric is in Section 5.2. Train FLOPs are computed from tokens seen via Eq. (2) in all rows. The equal-loss saving is read horizontally off the loss–FLOPs curves. At 500M the final gap is smaller than our measurement noise and changes sign between the two metrics, so we quote no equal-loss multiplier there; see the text and Section 5.2. [TODO: add 1B]

Scale	Arm	Raw tokens	Train FLOPs	Final eval	vs. base	Saving
50M	$k=1$	1.0B	3.50×10^{17}	4.437	–	
50M	$k=2$	2.0B	3.24×10^{17}	4.363	0.928×	1.48×
125M	$k=1$	2.5B	2.08×10^{18}	4.204	–	
125M	$k=2$	5.0B	1.94×10^{18}	4.171	0.932×	1.43×
250M	$k=1$	5.0B	8.34×10^{18}	3.414	–	
250M	$k=2$	10.0B	7.84×10^{18}	3.403	0.940×	1.31×
500M	$k=1$	10.0B	3.22×10^{19}	3.105	–	
500M	$k=2$	20.0B	3.05×10^{19}	3.107	0.946×	n/a

logged step gives 1.23× on the full-position metric; reading them one logging interval earlier gives 1.08×; and on the native metric the averaged curve never reaches the baseline’s final loss at all. A number that can be anything from 1.08× to “never” depending on which of two adjacent evaluation points we use is not measuring a compute saving, so we do not report one.

All the readings above come from native eval losses, and it is worth being precise about what they can support. The $k=1$ and $k=2$ native losses average over different sets of predictions (Section 3.6), so one arm’s loss-versus-compute curve is not measuring the same task as the other’s. Each curve is good evidence about its own family’s trajectory, but the equal-loss multipliers we read across curves are provisional until we repeat the same reading on the offset-ensemble metric. We have that metric at 125M and 500M, and at both scales it agrees with the direction of the native ordering (Section 5.2); the 50M and 250M pairs still need rescaling. The flipped native gap at 500M shows why the native metric cannot settle questions across k : on the comparable metric the ordering at that scale runs the other way, and both readings are small enough that neither decides anything.

Second, and subject to that caveat, the equal-loss multiplier is large at the three smallest scales but shrinks with every increase in model size: 1.48× at 50M, 1.43× at 125M, 1.31× at 250M, and at 500M the gap is too small to measure a multiplier from at all. Averaging is therefore not a fixed percentage off the training bill. The discount depends on scale, and over the range we cover it is shrinking, reaching our noise floor by 500M.

We deliberately do not extrapolate a parity scale from these points, in either direction. A decay that continues to zero and a decay that flattens just above parity are both consistent with what we measure, and such an extrapolation is easy to move. Fitting the multiplier linearly in $\log_{10} N$ over the first three scales reaches parity near 5×10^9 parameters under Eq. (2), but near 2×10^9 if the output projection is left out of that equation. A bookkeeping choice that changes no measured loss thus moves the projected crossover by more than a factor of two. The 500M point, which arrived after those fits were made, came in below what the three-point fit projected for it. That argues that the decay continues rather than flattens, but one unreplicated run at one scale cannot establish it. [TODO: add 1B point; fit $L(C)$ per family on the offset-ensemble metric and derive the crossover from that fit; seed replicates at 500M are needed before the noise-floor claim can be stated quantitatively]

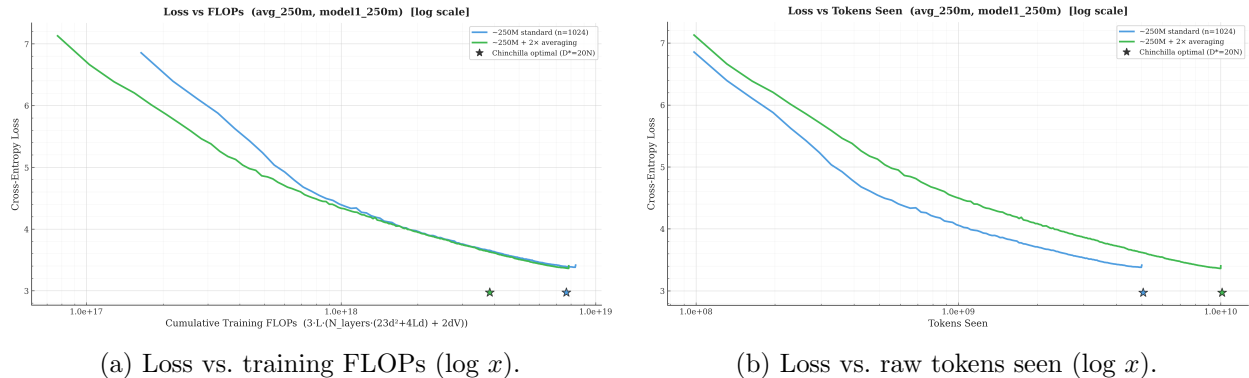


Figure 3: 250M comparison on native eval losses, both arms at their full budgets (5B and 10B raw tokens). The picture matches the smaller scales: the $k=2$ curve sits below the baseline at equal FLOPs throughout, and slightly above it on the raw-token axis. The gap at the end is visibly narrower here than at 125M, which is what the shrinking efficiency multiplier looks like.

Third, the raw-token view (Figure 2b) shows the mechanism directly. Per raw token the averaged model learns *slightly less* than the baseline, because compression really does lose information, but it processes each token at roughly half the compute, and the trade nets out positive. Token averaging does not make the model better per token; it makes tokens cheap enough that seeing twice as many wins.

5.2 Full-position evaluation: the ordering holds, with a shrinking margin

The curves above show native eval losses, which Section 3.6 argued do not compare like with like across k . Table 3 therefore rescores the final checkpoints with the offset-ensemble protocol: identical held-out sequences and identical target positions (3,409,119 predictions per model at 125M).

The two scales in the table are measured slightly differently, which matters when comparing their margins. The 125M rows come from rescoring the saved checkpoints offline with `eval_full_positions.py` over the full held-out set. The 500M rows come from the same construction run inside the training loop, which we added once random-offset training was in place so that later runs would not depend on keeping checkpoints; it runs the averaged model once per offset $o \in \{0, \dots, k-1\}$ and combines the resulting losses, weighted by how many predictions each contributes. The construction is the same either way, but the in-training version sees a smaller held-out sample, so the 500M margin carries correspondingly wider uncertainty.

At 125M the averaged model is better by 0.035 nats, a **3.4% perplexity improvement**, while having spent 0.93 $\times$ the training FLOPs. This is on a metric that scores every position exactly once from its full compressed prefix, so it answers the comparability objection directly rather than working around it.

At 500M the ordering is the same but the margin is not: 0.002 nats, a 0.2% perplexity improvement, at 0.946 $\times$ the training FLOPs. The sign and the magnitude deserve separate treatment. The sign is a real result, and it is the one the comparable metric gives: the averaged model is ahead, and it is ahead while having spent less compute, so nothing at this scale contradicts the claim that averaging buys loss per FLOP. The magnitude is no longer useful. A 0.002-nat margin from one unreplicated run is an order of magnitude below the 0.070-nat spread we measured at 50M from changing the training protocol alone (Section 5.4), and it is smaller than the gap between two adjacent evaluation points of the same run. We therefore read the 500M pair as showing that

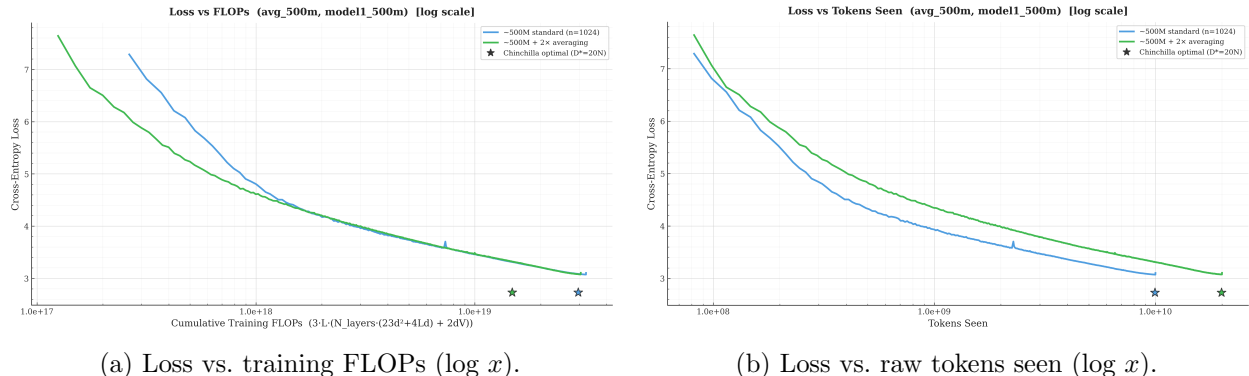


Figure 4: 500M comparison on native eval losses, both arms at their full budgets (10B and 20B raw tokens). The shrinking gap visible from 125M to 250M closes here: the two FLOPs-axis curves are indistinguishable over the final decade of training, and the final separation is a few thousandths of a nat, in favour of whichever arm the chosen metric favours. The raw-token panel still shows the mechanism intact, with the averaged model below the baseline in compute but above it per token. A single isolated spike in the baseline’s logged eval at step 69,000, which recovers within one logging interval, is excluded from the plotted curve (Appendix F).

the advantage has decayed into the noise, not as a 0.2% win. Deciding whether a small positive effect survives at this scale needs seed replicates, which we have not yet run. [TODO: 3 seeds at 500M with per-seed full-position NLL; report mean and standard deviation so the sign at this scale can be stated with a confidence interval rather than as a point estimate]

Splitting the 125M ensemble by offset confirms the argument of Section 3.6: offset 0 scores 4.145 and offset 1 scores 4.138, which is essentially equal. No offset is privileged. That is what random-offset training over packed data should produce, and it is what makes the ensemble a faithful full-sequence likelihood rather than one in-distribution pass mixed with $k-1$ out-of-distribution ones. Rotating the offset during generation relies on the same property (Appendix B). Taken together, random-offset training and offset-ensemble evaluation are what let patch-style training be a *final* operating point rather than a warm-up.

5.3 Where the gain comes from: cheaper data, not longer context

The gains of the more-data setting could come from either of two sources: more data per FLOP, or, in the more-context setting, more raw context behind each prediction. To separate them we trained four 50M models, all at about equal compute except the full-attention control, which is deliberately more expensive:

- the baseline ($L=1024$, 1024 tokens of raw context);
- more-context $k=2$ (`seq_len` 2048, $L=1024$, so $2\times$ the raw context at the same transformer cost);
- more-context $k=4$ (4096 tokens of raw context);
- a $k=1$ model with a genuinely 2048-token attention window, at $1.14\times$ the baseline’s cost.

Native losses are comparable only within a fixed k and grouping scheme, so every comparison below is within one method: each longer-context arm against the same- k arm without the longer context.

Table 3: Offset-ensemble full-position evaluation of the final more-data checkpoints. Within each scale, both rows average over the identical set of scored positions (every position $> k$ of each evaluation sequence), with the baseline restricted to exactly the positions the $k=2$ ensemble covers. The 125M rows are offline rescorings over the full held-out set; the 500M rows are the same construction computed in-training over a smaller sample. [TODO: fill 50M and 250M rows]

Scale	Model	Mean NLL (nats/token)	PPL	Coverage
125M	$k=1$ baseline (restricted)	4.177	65.1	all scored positions
125M	$k=2$, 2-offset ensemble	4.141	62.9	all scored positions
500M	$k=1$ baseline (restricted)	3.105	22.30	all scored positions
500M	$k=2$, 2-offset ensemble	3.102	22.25	all scored positions
50M	$k=1$ baseline (restricted)	[TODO:]		
50M	$k=2$, 2-offset ensemble	[TODO:]		
250M	$k=1$ baseline (restricted)	[TODO:]		
250M	$k=2$, 2-offset ensemble	[TODO:]		

Table 4: Context experiment at 50M, native eval losses. The Δ column compares each longer-context arm against the same- k arm without the longer context: $k=1$ full attention against the $k=1$ baseline; more-context $k=2$ against more-data $k=2$ at 4.363; more-context $k=4$ against more-data $k=4$ at 4.446. Losses in rows with different k are not comparable with each other.

Arm	Raw ctx	L	Raw tokens	Train FLOPs	Final eval (Δ within method)
$k=1$ baseline	1024	1024	1.00B	1.95×10^{17}	4.437
$k=1$ full attention	2048	2048	1.00B	2.45×10^{17}	4.651 (+0.214)
More context, $k=2$	2048	1024	2.00B	1.95×10^{17}	4.650 (+0.287)
More context, $k=4$	4096	1024	4.07B	1.99×10^{17}	5.174 (+0.728)

Table 4 and Figure 5 give the result, and it is the same in every comparison. **Longer raw context does not pay for itself at this scale, in either method.** Within $k=1$, widening the attention window from 1024 to 2048 costs +0.214 nats and $1.14\times$ the compute. Within $k=2$, spending the averaging saving on doubled raw context (4.650) instead of more data (4.363) costs +0.287 nats on the same prediction task. Within $k=4$ the penalty grows to +0.728 nats.

This is a property of the data and the scale, not of averaging. Most FineWeb documents are shorter than 1024 tokens, and packed sequences cross document boundaries without masking, so context past 1024 tokens is mostly made up of *other documents'* text. The more context we add, the more of it is irrelevant, which is why the penalty grows with the amount of extension. The consequence for the more-data setting is the point of the experiment: extra context here is worthless at best and harmful at worst, by every within-method reading, so the gains of the more-data setting are not a context effect. On packed web text, what token averaging buys is data per FLOP.

The experiment also answers the reverse question. When longer raw context is what you actually want, averaging is the cheaper way to get it. The two 2048-context arms finish at the same loss (4.6503 for $k=2$ averaging, 4.6505 for full attention), but the averaged route spends $1.14\times$ less training compute and holds half the KV cache and attention cost at inference for the same raw context. [TODO: report the full-position endpoints for the two 2048-context arms here, which is the comparable statement of this equality] This matters most on data where long context genuinely

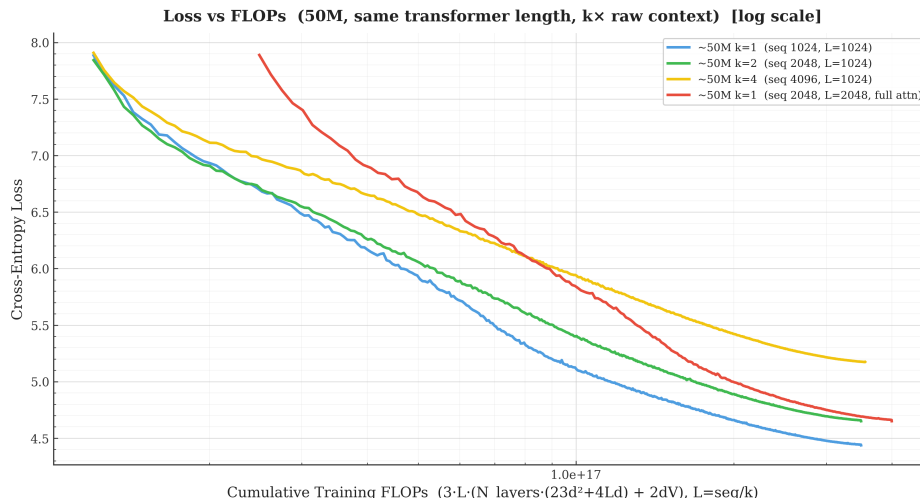


Figure 5: Context experiment at 50M: native eval loss vs. training FLOPs (log x). Curves with different k score different sets of positions and are overlaid only for orientation; the valid comparisons are within one method (see text). The two arms with 2048 tokens of raw context (green: $k=2$ averaging; red: full attention) end at almost the same native loss at very different cost, and $4\times$ context (yellow) is worse still. Early-training loss spikes are clipped at 8.0 for readability.

helps, such as long documents or code, which we leave to follow-up work. [TODO: loss-vs-position curves; document-length distribution of FineWeb; PG19/code long-context experiment]

5.4 Pooling functions: weighting by content and position beats the mean

Everything the transformer learns passes through the pooling function, so we compared the six functions of Section 3.7 at 50M in the more-data setting, under one matched protocol: identical architecture, data, batch size, update schedule, warmup, embedding tying, and evaluation, with a fixed token budget within each k (2.00B raw tokens at $k=2$, 4.07B at $k=4$). Within each k block the runs share the same k and, apart from the word-aligned one, the same grouping scheme, so their native losses are comparable with each other. Figure 6 shows all the curves, and Table 5 the final and quarter-budget losses.

One comparison does not hold. The word-aligned variant predicts a different distribution of positions (word starts only) at a slightly different realized compression, so its native loss is not measured on the same prediction task as the fixed-size variants. We report it as indicative and leave it out of the ordering (Appendix C).

One rule leads at both compression ratios. The pooler that weights tokens by content and position gives the lowest native loss at both $k=2$ and $k=4$, and it is the only rule whose standing holds up at both. The two fixed rules swap places: the plain mean is second best at $k=2$ and last at $k=4$, while the fixed recency weighting is last at $k=2$ and second best at $k=4$. At $k=2$ the learned pooler finishes 0.037 nats ahead of the mean, and it is ahead at every quarter-budget checkpoint. At $k=4$ its margin over the mean grows to 0.283 nats, which is 0.102 ahead of the next-best rule, the fixed recency weighting, and again it leads at every checkpoint. It costs $d_{\text{model}} + 1 + k$ parameters, which is negligible at any scale.

Every entry in Table 5 is a single run at one seed, so these are observed orderings rather than established ones. At $k=4$ the 0.283-nat margin is too large for seed noise to explain. At $k=2$

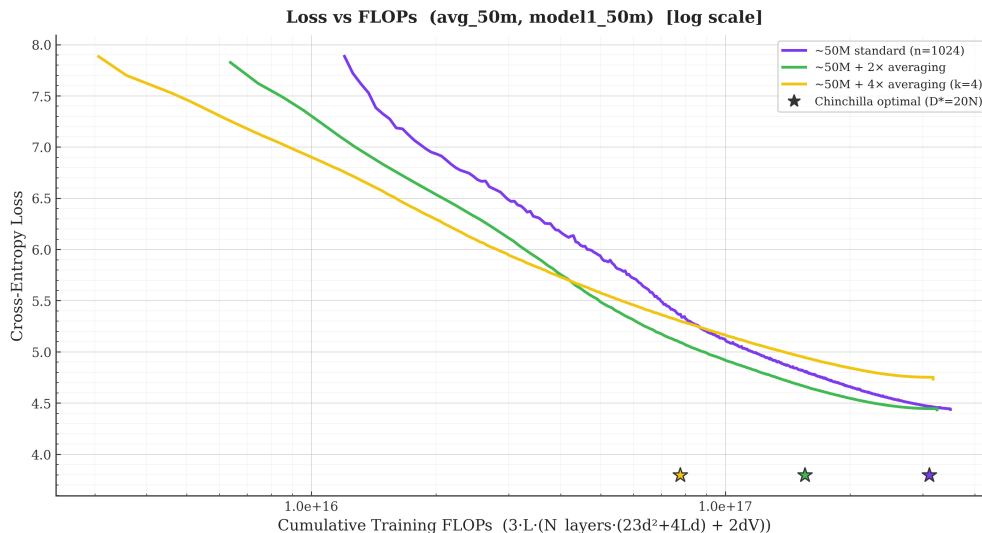


Figure 6: Pooling functions at 50M: native eval loss vs. recomputed training FLOPs (log x), with all $k=2$ and $k=4$ variants overlaid on the $k=1$ baseline and the original mean-pooling runs. Curves are comparable with each other only within one k block and one protocol; see text.

the 0.037-nat margin is not, and we do not yet have the replicates to bound it. Being ahead at every quarter-budget checkpoint is weaker evidence than independent seeds would be, because checkpoints within one run are strongly correlated. [TODO: SEED REPLICATES (planned; the single owner of every seed/variance claim in this paper, other seed notes point here). *Runs:* avg_50m_k2_learnable_pos and avg_50m_k2 (matched mean) at 2 additional seeds each, i.e. 4 runs of 2B raw tokens under the matched protocol (16,384 raw tokens/update, tied embeddings, 2k warmup), for $\sim 1.3 \times 10^{18}$ total FLOPs. Optionally add avg_50m_k4_learnable_pos and avg_50m_k4 at 1 additional seed each to confirm the $k=4$ margin, though 0.283 nats is unlikely to be seed noise. *Report:* per-variant mean and standard deviation of final native eval loss, and the seed spread of the learnable+pos minus mean difference at $k=2$. If that spread is comparable to 0.037 nats, downgrade the $k=2$ ordering to “indistinguishable” in the abstract, the Findings list, Table 5, and this section. *Then remove:* this paragraph’s hedging, the variance caveat in Table 5’s caption, and the seed note in Appendix E.]

What the mean seems to miss is position, not content. The remaining rows show where the win comes from. At $k=4$, the *fixed* recency weighting on its own recovers most of the gap to the leader (4.265 against 4.446 for the mean), while the scorer that sees only content recovers none of it (4.445, indistinguishable from the mean).

One confound has to be stated before reading that causally. At $k \geq 4$ we initialize the content-plus-position pooler’s position logits *at* the exponential recency rule, while the content-only scorer is zero-initialized and so starts at plain mean pooling. Part of the $k=4$ separation between the two learned poolers therefore comes from where they start rather than from what they can express, and a clean test would initialize both the same way. At $k=2$ both start uniform, so that comparison is free of the confound, and there the position term still helps. The fixed-weighting rows do not depend on initialization at all, and they point the same way.

The likely explanation is architectural. The content scorer looks at each embedding *on its own*, *before the network has any positional signal*, so it cannot express “the last token of the group matters

Table 5: Pooling functions at 50M in the more-data setting, under one matched protocol. Native eval CE (nats/token) at the nearest logged checkpoint to each quarter of the token budget. Runs are comparable within a k block; the word-aligned row is measured on a different prediction task (Appendix C). Every entry is a single run at one seed. [TODO: add variance bars once the seed replicates specified in Section 5.4 are available]

k	Pooling	25%	50%	75%	Final	Final PPL
2	learned: content + position	5.170	4.619	4.389	4.326	75.7
2	mean	5.174	4.649	4.425	4.363	78.5
2	learned: content only	5.222	4.675	4.443	4.383	80.1
2	overlapping groups ($w=4, s=2$)	5.460	4.820	4.562	4.492	89.3
2	word-aligned (task differs)	5.413	4.872	4.640	4.577	97.2
2	fixed recency	5.439	4.962	4.707	4.617	101.2
4	learned: content + position	4.951	4.463	4.224	4.163	64.3
4	fixed recency	5.041	4.569	4.330	4.265	71.1
4	learned: content only	5.315	4.753	4.512	4.445	85.2
4	mean	5.326	4.756	4.505	4.446	85.2

most”. That is exactly the prior that pays off when four tokens collapse into one vector, since the group’s last token is the one next to the prediction target. At $k=2$ the same prior, applied as an aggressive fixed rule, actively hurts: it gives 4.617, the worst result we measured. With only two tokens per group both carry useful signal, and systematically down-weighting the first throws that away. The content-plus-position pooler resolves the tension by *learning* the positional profile for each compression ratio, and content information then adds a further gain on top at both values of k . As for the structural alternatives, overlapping groups (which remove hard boundary cuts entirely) and word-aligned groups both do worse than the plain mean, so neither boundary effects nor linguistically meaningful grouping is the main thing at work.

Why the matched protocol matters. Training details move the results by more than the pooling functions do. Mean pooling trained under our earlier protocol ($\sim 8\times$ fewer optimizer updates for the same token budget, and correspondingly different warmup exposure) ends 0.070 nats worse at $k=2$ and 0.288 nats worse at $k=4$ than the same model under the matched protocol. The update schedule alone therefore moves final losses by more than most of the gaps in Table 5. That is why every pooling function in this section is trained under one identical protocol, and why we keep the older mean-pooling runs in Appendix E as references only and never use them as controls. [TODO: offset-ensemble rescaling of the pooling winners; learned weight profiles (Appendix D); seeds are specified in the seed replicate note above]

6 Discussion and Limitations

Scale. The efficiency multipliers we can measure come from the 50M–250M range, a factor of five in parameters, and they shrink across all of it. The 500M pair extends the range to a factor of ten, and it is consistent with a decay that continues: the comparable-metric gap there is 0.002 nats, which is too small for us to tell a small positive effect from noise, and the equal-loss multiplier can no longer be measured stably at all.

We do not quote a parity scale in either direction. Fitting the multiplier linearly in $\log_{10} N$ over the first three scales gives parity near 5×10^9 parameters under Eq. (2), and near 2×10^9 if the

output projection is dropped from the FLOPs accounting; a bookkeeping choice that changes no measured loss therefore moves the answer by more than a factor of two. The 500M point also came in below what that fit projected for it, which is a further reason not to read such extrapolations literally. The honest summary is that the advantage is real and sizeable below 250M, has decayed to within noise by 500M, and that we cannot yet say whether it settles at zero or at some small positive value. This is the largest open question in the paper. Answering it needs two things we have not done: seed replicates at 500M, so the sign at that scale gets a confidence interval, and the 1B pair, which extends the range by another factor of two.

Data logistics are part of the difficulty. The $k=2$ arm consumes twice the raw tokens, and the 250M $k=2$ run already exhausts FineWeb sample-10BT, so the 500M pair and everything above it draws from `sample-100BT`. We change the slice for both arms of a scale, so no within-scale comparison is affected, but the 500M and 1B scale points are not drawn from exactly the same subsample as the smaller ones, and cross-scale fits should be read with that in mind. [TODO: 1B results; seed replicates at 500M]

Tokenizer granularity. Averaging two tokens looks a little like using a tokenizer whose tokens are twice as long. The two are not the same, because averaging keeps the fine vocabulary and composes embeddings on the fly, but the direct control is a coarser-tokenizer baseline compared in bits-per-byte. [TODO: coarser-tokenizer baseline in bits-per-byte]

Generation and downstream evaluation. Everything we report is language-modeling loss. Rotating-offset decoding (Appendix B) gives token-by-token generation at about the cost of an ordinary decoder, and downstream zero-shot results (LAMBADA, HellaSwag, PIQA, ARC-E) are the natural complement to the loss numbers. A small local decoder that emits all k tokens of a group, in the style of Block Transformer, would give native generation and is a further extension. [TODO: rotating-offset decoding benchmark; downstream evals]

How far the context result generalizes. Our finding that longer raw context hurts is a statement about packed FineWeb at these scales, where most documents are shorter than 1024 tokens and packing crosses document boundaries without masking. It is not a claim about long-document corpora, code, or masked packing. On those, the fact that averaging buys context more cheaply than full attention could turn into a genuine long-context win. We think this is the most promising unexplored part of the work.

7 Conclusion

[TODO: Write after the 1B endpoints and the remaining full-position numbers land. Skeleton: (i) token averaging turns cheap data into lower loss at fixed compute, giving 3.4% lower perplexity at $0.93\times$ the FLOPs on the comparable metric at 125M and equal-loss compute savings of $1.48\times$ at 50M, $1.43\times$ at 125M and $1.31\times$ at 250M, but the multiplier shrinks with scale and by 500M the gap is 0.002 nats, within noise, so the honest headline is a method that works well below 250M and whose benefit at larger scale is unresolved pending 500M replicates and the 1B pair; (ii) random-offset training and offset-ensemble evaluation solve the comparability problem and let patch-style operation be a final operating point, with rotating-offset decoding for generation; (iii) longer raw context hurts on packed web text, so the gain is cheaper data, while averaging is the cheaper way to buy context when context is what you want; (iv) weighting tokens by content and

position is the best pooling rule at both compression ratios, and position is the ingredient the mean discards. Close with the scaling question as the decisive open item.]

References

[TODO: complete bibliography when Related Work is written]

Hoffmann, J., Borgeaud, S., Mensch, A., Buchatskaya, E., Cai, T., Rutherford, E., de Las Casas, D., Hendricks, L. A., Welbl, J., Clark, A., Hennigan, T., Noland, E., Millican, K., van den Driessche, G., Damoc, B., Guy, A., Osindero, S., Simonyan, K., Elsen, E., Rae, J. W., Vinyals, O., & Sifre, L. (2022). Training compute-optimal large language models. *arXiv preprint arXiv:2203.15556*.

A FLOPs Derivation

Per layer and per transformer position, counting multiply-accumulates as 2 FLOPs, forward pass:

- Attention projections: Q, K, V , and output, i.e. four $d \times d$ matmuls: $4 \times 2d^2 = 8d^2$.
- SwiGLU FFN with `ff_multiplier` 2.5: the combined implementation projects $d \rightarrow 5d$ for up and gate ($2 \cdot 5d^2 = 10d^2$), then projects down $2.5d \rightarrow d$ ($2 \times 2.5d^2 = 5d^2$): total $15d^2$. (A standard $4 \times$ FFN would give $16d^2$; our architecture’s SwiGLU at 2.5 gives $15d^2$, and we use the exact value.)
- Attention scores and value aggregation: $2 \cdot 2Ld = 4Ld$ per position at sequence length L (causal masking halves the effective term; we follow the standard convention of quoting the full-matrix count and apply it uniformly to all arms, so ratios are unaffected). [TODO: state the causal-mask convention once and verify consistency with plotting code]
- Output projection: one $d \rightarrow V$ matmul per scored position, $2dV$, with $V = 50,304$. Each transformer position emits exactly one prediction in both arms, so this term is proportional to L and not to `seq_len`.

Total forward: $n_{\text{layers}}(23d^2 + 4Ld) + 2dV$ per position. Training multiplies by 3 (forward + backward). A sequence of `seq_len` raw tokens occupies $L = \text{seq_len}/k$ positions, which gives Eq. (2). Every plot and table in this paper computes FLOPs from `tokens_seen` through that equation.

The output projection is a large share of the total at these scales rather than a correction term, so we tabulate it explicitly. As a fraction of the baseline arm’s total training FLOPs it is 44.2% at 50M ($d=512$), 27.8% at 125M ($d=768$) and 18.5% at 250M ($d=1024$), falling as d grows towards V . Both arms score one token per transformer position, and the equal-compute design gives them equal position counts, so in absolute terms this term contributes *equally* to both. Including it therefore moves every efficiency ratio towards 1: the final cost ratios are 0.928/0.932/0.940 with the head and would be 0.871/0.906/0.926 without it, and the equal-loss multipliers are 1.48/1.43/1.31 with the head and 1.58/1.47/1.33 without. The head-exclusive figures do not describe total training cost, so we do not report them as though they did.

We exclude input embedding lookups on both arms. The lookup is a gather rather than a matmul, and the averaged arm performs $k \times$ as many of them per transformer position, so leaving it out is the conservative choice for the method.

B Offset-Ensemble Evaluation: Details

For a k -averaged model and evaluation sequence $t_{1..T}$, offset $o \in \{0, \dots, k-1\}$ discards the first o tokens, forms $n_o = \lfloor (T - o)/k \rfloor$ groups, embeds and pools them, and runs the transformer on the first $n_o - 1$ compressed positions. The prediction at compressed position j (1-indexed) is scored against the raw token at position $o + jk + 1$. Across offsets, the sets of predicted positions never overlap, and together they cover every position $> k$ up to edge effects at the end of the sequence, each predicted exactly once from its full compressed prefix. We compute mean NLL by pooling all scored predictions across offsets and sequences. For the restricted baseline, we gather the $k=1$ model’s per-position NLLs on exactly this set of positions. The total cost is k forward passes at length $\approx T/k$, which is about one baseline pass at length T ; the attention part is in fact cheaper by a factor k .

Generation rotates the offset. To emit the token at raw position $p+1$, the decoder uses the offset $o \equiv p \bmod k$, which is the one whose group boundaries end exactly at p . Keeping k caches, one per offset, amortizes to about one ordinary decoder’s cost, because each cache is $1/k$ the length. [TODO: benchmark rotating-offset decoding; measure downstream tasks]

C Word-Aligned Windows: Algorithm and Caveats

A token begins a word if its string form starts with the BPE whitespace marker (a leading $\mathbf{G}$ in GPT-NeoX byte-level BPE) or a newline marker, or if it is a special token. We build a boolean lookup table over the padded vocabulary once from the tokenizer. Grouping is greedy within each sequence: accumulate tokens into the current group; close the group when it holds $\geq k$ tokens *and* the next token starts a new word, or unconditionally at $2k$ tokens so that an unusually long word cannot grow a group without bound; then drop any trailing partial group. Groups therefore span 2–4 tokens at $k=2$ and never end mid-word unless a word exceeds $2k$ tokens, and the realized compression on FineWeb is $\sim 2.1\times$ at $k=2$. Within a batch we truncate all sequences to the smallest group count. The grouping loop costs $\sim 1\text{ms}$ per batch at $B=16$, $T=1024$, which is negligible. Targets are the first token of the next group, which for this variant is always the start of a word.

Two caveats follow. First, the realized compression exceeds the nominal $2\times$, so the FLOPs accounting has to use the realized ratio; the transformer sees $\sim 5\%$ fewer positions per raw token than uniform $k=2$. Second, the distribution of predicted positions differs, since only word starts are predicted, so this variant’s native eval loss is not measured on the same prediction task as the fixed-size variants. The offset-ensemble protocol does not directly apply to variable-size groups either. A bits-per-byte comparison or a fixed-position rescoring is needed before making strong claims, so we present the word-aligned row as indicative only. [TODO: bits-per-byte rescoring for the word variant]

D Pooling Modules

Learned from content only. Shared scorer $s_i = w^\top e_i + b$ with $w \in \mathbb{R}^{d_{\text{model}}}$ and $b \in \mathbb{R}$, zero-initialized so that $\alpha_i = 1/k$ exactly at initialization and training starts at mean pooling; $\alpha = \text{softmax}(s_{1..k})$ within each group; $\bar{e} = \sum \alpha_i e_i$. Parameters: $d_{\text{model}} + 1 = 513$. It trains jointly with the language model under the ordinary objective, with no auxiliary reconstruction loss.

Learned from content and position. $s_i = w^\top e_i + b + p_i$, with $p \in \mathbb{R}^k$ a learned bias per slot of the group. Initialization: $p = 0$ (uniform) at $k=2$, where the recency prior hurt, and $p_i \propto$ the

exponential profile with last-to-first ratio 20 at $k=4$, where it helped. Parameters: $d_{\text{model}} + 1 + k$. Both terms train end-to-end. Inspecting the learned α_i by position and token type is part of the planned analysis. [TODO: report learned weight profiles]

Fixed weighting schemes. $w_i \propto \exp(\alpha_i)$ with $\alpha = \ln(20)/(k-1)$, which is the exponential rule we trained, plus linear, Gaussian, and triangular alternatives that we implemented but did not train at scale. All schemes are L_1 -normalized.

E Full Run Table

Table 6: All runs referenced in this paper, with canonical result files. “Protocol” distinguishes the matched protocol used for all canonical runs (16,384 raw tokens per update, tied embeddings, 2k-step warmup) from the original protocol of earlier runs, which are retained as references only. All models use tied embeddings. Every row is a single run at one seed. [TODO: add exact batch size, world size, and step counts per run; add a seed column once the replicates specified in Section 5.4 are available; add 1B rows as they complete.]

Run	k	Raw tokens	Final eval	Protocol	Notes
50M $k=1$ baseline	1	1.0B	4.437	matched	canonical baseline
50M $k=2$ mean	2	2.0B	4.363	matched	canonical
50M $k=4$ mean	4	4.07B	4.446	matched	canonical
50M $k=1$, 2048 full attn	1	1.0B	4.651	original	context control
50M $k=2$, seq 2048 (more context)	2	2.0B	4.650	original	context arm
50M $k=4$, seq 4096 (more context)	4	4.07B	5.174	original	context arm
125M $k=1$	1	2.5B	4.204	original	
125M $k=2$	2	5.0B	4.171	original	
250M $k=1$	1	5.0B	3.414	original	32,768 raw tok/update
250M $k=2$	2	10.0B	3.403	original	32,768 raw tok/update
500M $k=1$	1	10.0B	3.105	original	32,768 raw tok/update; 100BT slice
500M $k=2$	2	20.0B	3.107	original	32,768 raw tok/update; 100BT slice
50M $k=2$ learned: content + position	2	2.0B	4.326	matched	pooling comparison
50M $k=2$ learned: content only	2	2.0B	4.383	matched	pooling comparison
50M $k=2$ overlapping groups $w4s2$	2	2.0B	4.492	matched	pooling comparison
50M $k=2$ word-aligned	2	2.0B	4.577	matched	metric caveats
50M $k=2$ fixed recency	2	2.0B	4.617	matched	pooling comparison
50M $k=4$ learned: content + position	4	4.07B	4.163	matched	pooling comparison
50M $k=4$ fixed recency	4	4.07B	4.265	matched	pooling comparison
50M $k=4$ learned: content only	4	4.07B	4.445	matched	pooling comparison
50M $k=2$ mean (old protocol)	2	2.0B	4.433	original	reference only
50M $k=4$ mean (old protocol)	4	4.07B	4.734	original	reference only

F Training Instabilities and Excluded Runs

For completeness and reproducibility we document the detours.

1. **RoPE bug.** An early implementation bug in the rotary embedding invalidated the first tied-embedding run and several early runs across values of k . All affected logs are quarantined (`*_rope_bug` suffixes in the results tree) and no number in this draft comes from them.
2. **50M baseline instability.** The original 50M $k=1$ run suffered an early loss explosion and converged ~ 1 nat too high. A clean rerun, the 4.437 baseline used throughout, matches the

cross-scale pattern. All runs at this scale show a large transient loss spike in the first $\sim 2k$ steps that recovers on its own; it is an initialization artifact, and we clip it in plots.

3. **The seq_len accident.** The first $k=2$ runs were launched with the baseline’s raw `seq_len` rather than the doubled value originally intended. The accidental setting holds raw context fixed and isolates the data-per-FLOP effect, and it became the paper’s main setting (more data); the originally intended setting became the more-context ablation.
4. **Duplicated ablation log.** The $k=2$ learnable run’s CSV contains two interleaved logging streams and one malformed line (from a resumed job double-writing); the two recorded endpoints agree to 0.004 nats (4.379 vs. 4.383) and the file is cleaned deterministically before plotting.
5. **500M baseline eval spike.** The 500M $k=1$ log contains a single isolated evaluation point at step 69,000 (2.26B tokens) where train and eval loss jump from ~ 3.58 to 7.25 and recover fully by the next logged step 500 steps later. It is a one-interval transient, not a divergence, and the run completes normally; it is excluded from the plotted curve by a filter that drops any point exceeding both its neighbours by more than 40%, and it affects no reported endpoint.
6. **500M $k=2$ resume and duplicated steps.** The 500M $k=2$ run was interrupted at $\sim 98\%$ of its budget by a full disk, which truncated the in-progress checkpoint. Training resumed from the last intact checkpoint (step 550,000), so the appended log contains 100 duplicated logging steps covering 550,500–600,000. We deduplicate by step, retaining the post-resume values, which is the trajectory that actually produced the final weights. Optimizer and scheduler state are checkpointed alongside the model, so the resumed segment continues the same cosine schedule rather than restarting it. Checkpoint writes are now atomic and rotated to prevent recurrence.

G Hyperparameters

Table 7: Shared hyperparameters. [TODO: fill exact per-run batch sizes and world sizes from run logs.]

Optimizer	AdamW
LR schedule	cosine, decay to 10% of peak over total steps
Warmup	2,000 steps
Peak LR	$2.0/1.6/1.4/1.2 \times 10^{-4}$ (50M/125M/250M/500M)
Precision	bfloat16 autocast
Raw <code>seq_len</code>	1024 (more data; more context as stated)
Tokenizer	EleutherAI Pythia-70m (GPT-NeoX BPE), vocab 50,257
Vocabulary padding	50,304
FFN	SwiGLU, expansion 2.5
Positional encoding	RoPE (applied to compressed positions)
Chinchilla reference constants	$A=406.4$, $\alpha=0.3392$, $B=410.7$, $\beta=0.2849$, $E=1.6934$